Mini Project Report On

Nonlinear Dynamics in Neural Networks

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science & Engineering

By

Diya Jothish (U2303084)

Gayathri Binoj (U2303102)

Maria Elsha Thomas (U2303144)

Under the guidance of

Dr. Sminu Izudheen

Department of Computer Science & Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Affiliated to APJ Abdul Kalam Technological University)
Rajagiri Valley, Kakkanad, Kochi, 682039

March 2026

CERTIFICATE

This is to certify that the mini project report entitled ***"Nonlinear Dynamics in Neural Networks"*** *is a bonafide record of the work done by* ***Diya Jothish (U2303084)***, ***Gayathri Binoj (U2303102)*** *and* ***Maria Elsha Thomas (U2303144)***, *submitted to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

Dr. Sminu Izudheen
Professor
Dept. of CSE
RSET

Ms. Seema Safar
Assistant Professor
Dept. of CSE
RSET

Dr. Preetha K G
Professor
Dept. of CSE
RSET

ACKNOWLEDGEMENTS

We wish to express our sincere gratitude towards Rev. Dr. Jaison Paul Mulerikkal CMI, Principal of RSET, and Dr. Preetha K.G., Head of the Department of Computer Science and Engineering for providing us with the opportunity to undertake our mini project, "Nonlinear Dynamics in Neural Networks".

We are highly indebted to our project coordinators, Ms. Seema Safar, Assistant Professor, Department of Computer Science and Engineering, and Ms. Sherine Sebastian, Assistant Professor, Department of Computer Science and Engineering, for their valuable support.

It is indeed our pleasure and a moment of satisfaction for us to express our sincere gratitude to our project guide **Dr. Sminu Izudheen** for her patience and all the priceless advice and wisdom she has shared with us.

Last but not the least, we would like to express our sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

Diya Jothish

Gayathri Binoj

Maria Elsha Thomas

Abstract

This project focuses on modeling and analyzing the nonlinear and time-varying behaviour of large-scale brain networks using resting-state functional Magnetic Resonance Imaging (fMRI) data. The work begins with the extraction of Blood-Oxygen-Level-Dependant (BOLD) time-series signals from predefined regions of interests (ROIs) after standard preprocessing steps. Functional connectivity matrices are then constructed to represent the interactions between the different regions in the brain, showing the brain as a network of nodes and edges, using graph theory concepts. Sliding window techniques will be utilized to analyze the changes in functional connectivity over time, which helps understand the dynamic behavior of the brain. The resulting static and dynamic network representations are visualized and analyzed to point out the drawbacks of purely linear, static approaches in the interpretation of brain connectivity, which is highly dynamic in nature. This project provides a foundational framework for studying nonlinear brain dynamics and can be extended to advanced analysis and neurological disorder studies in the future.

Contents

Acknowledgements **i**

Abstract **ii**

List of Figures **v**

List of Tables **vi**

1 Introduction **1**

1.1 Background 1

1.2 Problem Definition 2

1.3 Scope and Motivation 2

1.4 Objectives 3

1.5 Challenges 4

1.6 Assumptions 4

1.7 Societal / Industrial Relevance 5

1.8 Organization of the Report 6

2 Software Requirements Specification **7**

2.1 Introduction 7

2.2 Overall Description 7

2.3 External Interface Requirements 8

2.4 System Features 10

2.5 Other Nonfunctional Requirements 11

3 System Architecture and Design **13**

3.1 System Overview 13

3.2 Architectural Design 15

3.3 Dataset identified 16

3.4 Proposed Methodology/Algorithms 17
3.5 User Interface Design 21
3.6 Database Design 22
3.7 Description of Implementation Strategies 25
3.8 Module Division 26
3.9 Work Schedule - Gantt Chart 28

4 Results and Discussions **29**
4.1 Overview 29
4.2 Testing 29
4.3 Quantitative Results 30
4.3.1 ROI Flexibility 31
4.3.2 Graph-Theoretic Measures 31
4.3.3 Node Strength 32
4.4 Graphical Analysis 32
4.5 Discussion 35

5 Conclusion **36**
5.1 Conclusion 36
5.2 Future Scope 37

Appendix A: Presentation **39**

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes **54**

Appendix C: CO-PO-PSO Mapping **59**

List of Figures

3.1 Architectural Diagram 13
3.2 Flow Chart 15
3.3 Use Case Diagram 16
3.4 User Interface Design 21
3.5 Gantt Chart 28

4.1 Static Functional Connectivity Matrices for Healthy and Parkinson's Subjects 30
4.2 Functional Connectivity Networks Thresholded at 0.5 30
4.3 Average ROI Switch Counts Comparison Between Healthy and Parkinson's Groups 31
4.4 Clustering Coefficient Over Time Across Sliding Windows 32
4.5 Global Efficiency Over Time Across Sliding Windows 33
4.6 Modularity Over Time Across Sliding Windows 33
4.7 Characteristic Path Length Over Time Across Sliding Windows 34
4.8 Group-Averaged Recurrence Plots for Healthy and Parkinson's Subjects . . 34

List of Tables

4.1 Group-Averaged Graph-Theoretic Measures 32

Chapter 1

Introduction

1.1 Background

The human brain comprises billions of neurons organized into specialized regions that communicate through intricate networks. Resting-state functional Magnetic Resonance Imaging (rs-fMRI) has emerged as a powerful non-invasive tool for studying these networks by measuring the Blood Oxygen Level Dependent (BOLD) signal — a proxy for neural activity based on changes in blood oxygenation [8]. By analyzing the temporal correlations between BOLD signals across brain regions, researchers can map functional connectivity — the statistical dependencies between spatially distinct regions that reflect coordinated neural activity [2].

Traditionally, functional connectivity has been studied using static methods, which compute a single correlation matrix averaged over the entire scan duration. While this has yielded significant insights into large-scale brain network organization, it fundamentally assumes that connectivity is stationary over time. However, growing evidence suggests that the brain is a highly dynamic system whose connectivity patterns fluctuate [1] on the order of seconds to minutes, even during rest. These fluctuations are not random noise — they reflect meaningful changes in functional organization that are directly relevant to cognition, behavior, and neurological disease.

Parkinson's disease is a progressive neurodegenerative disorder characterized by the loss of dopamine-producing neurons in the substantia nigra, disrupting motor circuits and producing the hallmark symptoms of tremor, rigidity, and bradykinesia. The basal ganglia — comprising the Caudate, Putamen, Pallidum, and their connections to the Thalamus and cortex — form the core of the affected circuitry, but the disease's effects propagate

through functional networks to cortical and limbic regions as well. Recent advances in dynamic functional connectivity analysis have opened new avenues for understanding how Parkinson's disease alters not just the average organization of brain networks, but their temporal dynamics — motivating the sliding window [3] and graph-theoretic approach taken in this project.

1.2 Problem Definition

Static fMRI connectivity analysis captures only the time-averaged functional organization of the brain, obscuring the dynamic fluctuations that may differentiate healthy neural function from the disrupted network dynamics characteristic of Parkinson's disease. This project aims to model and quantify the dynamic functional connectivity of the brain using resting-state fMRI data, comparing the temporal organization of brain networks in healthy subjects and Parkinson's disease patients to identify disease-specific patterns of dynamic dysregulation.

1.3 Scope and Motivation

This project focuses on the analysis of resting-state fMRI data from the Taowu dataset, comprising 20 healthy control subjects and 20 Parkinson's disease patients. The analysis is restricted to 24 brain regions of interest known to be implicated in Parkinson's disease, spanning the basal ganglia, limbic system, and motor cortex. The scope encompasses the full pipeline from data preprocessing and BOLD signal extraction to static connectivity analysis, dynamic sliding window analysis [1] [2], spectral graph partitioning, graph-theoretic characterization, and recurrence analysis [4]. The project does not extend to clinical diagnosis or treatment recommendation, but rather provides a computational framework for understanding network-level differences between the two groups.

The motivation for this work arises from the fundamental limitation of static connectivity analysis in capturing the brain's inherently dynamic nature. Parkinson's disease is known to disrupt the temporal organization of motor and cognitive circuits, yet conventional fMRI studies often reduce this rich temporal information to a single static snapshot. By applying dynamic connectivity analysis, this project seeks to reveal aspects of Parkin-

son's pathology that are invisible to static methods — specifically, whether the disease alters not just which brain regions are connected, but how stably and predictably those connections are maintained over time. Understanding these dynamic alterations could contribute to the development of more sensitive biomarkers for disease detection, progression monitoring, and therapeutic evaluation.

1.4 Objectives

- To preprocess resting-state fMRI data and extract BOLD time series signals from 24 Parkinson's disease-relevant regions of interest using a standard brain atlas.
- To compute static functional connectivity matrices and characterize network topology using graph-theoretic measures including degree centrality, clustering coefficient, and global efficiency.
- To implement a Gaussian-tapered sliding window approach for dynamic functional connectivity analysis, generating time-varying connectivity matrices across all subjects.
- To apply spectral graph partitioning to each window's connectivity matrix to identify dynamic functional community structure and quantify regional flexibility through ROI switch counts.
- To compute window-level graph-theoretic measures including clustering coefficient, path length, global efficiency, modularity, and node strength, and compare their temporal evolution between healthy and Parkinson's groups.
- To perform recurrence analysis on group-averaged connectivity trajectories to characterize the temporal structure of brain state dynamics in both groups.
- To synthesize findings from static and dynamic analyses to draw comprehensive conclusions about how Parkinson's disease disrupts the brain's functional network organization over time.

1.5 Challenges

The primary technical challenge in this project is the high computational cost of processing full-length fMRI scans for 40 subjects, as each subject requires BOLD extraction, sliding window computation across hundreds of windows, spectral partitioning, and graph-theoretic measure computation — all of which are computationally intensive operations. A further challenge arises from inter-subject variability in scan duration and brain coverage, as some subjects' fMRI scans have slightly different fields of view, causing certain atlas regions to fall outside the imaging volume and requiring careful handling to prevent missing data from skewing group averages. Additionally, the sign ambiguity of eigenvectors in spectral partitioning means that community labels are not directly comparable across windows or subjects without alignment, requiring careful interpretation of switch counts as a measure of change rather than absolute community identity.

1.6 Assumptions

1. The resting-state fMRI data is assumed to adequately capture intrinsic brain activity in the absence of a task, and that subjects remained still and awake during the scan.

2. The AAL (Automated Anatomical Labeling) atlas is assumed to provide a sufficiently accurate parcellation of the 24 regions of interest for both healthy and Parkinson's subjects, despite potential individual differences in brain anatomy.

3. A sliding window of 20 timepoints with a step size of 1 is assumed to provide an adequate temporal resolution for capturing meaningful fluctuations in functional connectivity without introducing excessive noise from overly short windows.

4. Group-level averages across 20 subjects per group are assumed to be sufficiently representative of the broader healthy and Parkinson's populations for the purposes of this analysis.

1.7 Societal / Industrial Relevance

From a societal perspective, Parkinson's disease affects over 10 million people worldwide and its prevalence is expected to increase significantly with aging populations. Early and accurate detection of disease-related brain network changes could substantially improve patient outcomes by enabling earlier intervention and more targeted treatment. The dynamic connectivity framework developed in this project could contribute to the development of imaging-based biomarkers for Parkinson's disease that are more sensitive than current static connectivity measures, potentially supporting earlier diagnosis before motor symptoms become severe.

From an industrial and clinical perspective, the computational pipeline developed here has direct applicability in neuroimaging research tools and clinical decision support systems. The graph-theoretic and dynamic analysis methods employed are generalizable beyond Parkinson's disease to other neurological conditions including Alzheimer's disease, epilepsy, schizophrenia, and major depressive disorder, all of which have been associated with alterations in functional network dynamics. As fMRI technology becomes more accessible and datasets grow larger, automated pipelines like this one that can process large cohorts and generate quantitative, interpretable reports will become increasingly valuable in both research and clinical settings.

1.8 Organization of the Report

The report is organized into five chapters as follows.

Chapter 1 presents the introduction to the project, covering the background of functional brain connectivity analysis and Parkinson's disease, the problem definition, scope and motivation, objectives, challenges, assumptions, societal and industrial relevance, and the organization of the report.

Chapter 2 presents the Software Requirements Specification, describing the overall system, external interface requirements including hardware, software, and data input specifications, system features corresponding to each functional module of the pipeline, and non-functional requirements including performance, reliability, portability, and maintainability.

Chapter 3 presents the System Architecture and Design, providing a detailed system overview of the full processing pipeline, architectural and use case diagrams, a description of the Taowu dataset, the proposed methodology and algorithms including sliding window connectivity, spectral graph partitioning, ROI switch computation, graph-theoretic measures, and recurrence analysis, the user interface design, the file-based database design, implementation strategies with code snippets, and the module division among team members.

Chapter 4 presents the Results and Discussions, describing the overall results achieved through the analysis, the testing and validation of the pipeline, quantitative results including ROI switch counts, graph-theoretic measures, and node strength comparisons, graphical analysis of the dynamic connectivity findings including graph measure time series and recurrence plots, and a discussion summarizing and interpreting the findings in the context of Parkinson's disease pathology.

Chapter 5 presents the Conclusion, summarizing the key findings from both static and dynamic analyses, discussing the complementary insights provided by the two approaches, and outlining potential directions for future work including larger datasets, alternative dynamic connectivity methods, formal statistical testing, and extension to other neurodegenerative conditions.

Chapter 2

Software Requirements Specification

2.1 Introduction

This Software Requirements Specification describes the functional and non-functional requirements for a computational neuroimaging analysis tool designed to model and quantify dynamic functional connectivity in resting-state fMRI data. The system processes fMRI scans from healthy subjects and Parkinson's disease patients, extracts BOLD time series signals, performs both static and dynamic connectivity analyses, and generates a comprehensive PDF report of findings. The intended users are neuroscience researchers and students with working knowledge of neuroimaging concepts and Python-based computational tools. This document serves as a reference for understanding the capabilities, constraints, and interface requirements of the system.

2.2 Overall Description

Product Perspective

The system is a standalone command-line analysis pipeline built in Python. It does not form part of a larger commercial software system but is designed to be modular and extensible, with clearly separated source modules for data loading, preprocessing, connectivity analysis, dynamic analysis, visualization, and report generation. It takes raw fMRI data as input and produces quantitative results, visualizations, animations, and a structured PDF report as output.

Product Functions

The system performs the following high-level functions. It loads and preprocesses resting-state fMRI data using standard neuroimaging pipelines. It extracts BOLD time series

from 24 Parkinson's disease-relevant regions of interest using the AAL atlas. It computes static functional connectivity matrices and visualizes them as heatmaps, network graphs, and connectome plots. It implements a Gaussian-tapered sliding window to generate time-varying connectivity matrices. It applies spectral graph partitioning to identify dynamic functional communities and compute ROI switch counts. It computes graph-theoretic measures per window including clustering coefficient, path length, global efficiency, modularity, and node strength. It generates recurrence plots from group-averaged connectivity trajectories. It produces animated visualizations of dynamic connectivity and network reorganization. It generates a structured PDF report summarizing all findings with quantitative comparisons and interpretations.

User Characteristics

The primary user is assumed to have familiarity with Python, basic neuroimaging concepts including fMRI and BOLD signals, and an understanding of functional connectivity analysis. No graphical user interface is provided — interaction is through a terminal-based menu system.

Constraints

The system requires significant computational resources due to the processing of high-dimensional fMRI data across multiple subjects. Processing all 40 subjects for dynamic analysis is time-intensive and may take several hours depending on hardware. The system is currently designed for the Taowu dataset structure and the AAL atlas, and adaptation to other datasets or atlases would require modifications to the data loading and preprocessing modules.

2.3 External Interface Requirements

Hardware Requirements

A minimum of 8GB RAM is recommended for processing full-length fMRI scans, with 16GB or more recommended for group-level dynamic analysis across all 40 subjects. A multi-core processor is beneficial as several underlying libraries support parallel computation. Sufficient disk space — at least 10GB — is required to store the fMRI dataset,

intermediate outputs, animations, and the generated report.

Software Requirements The system requires Python 3.8 or higher. The following external libraries are required:

- nilearn — for fMRI data loading, preprocessing, atlas handling, and neuroimaging visualization
- numpy — for numerical array operations and matrix computations
- scipy — for statistical functions and signal processing
- networkx — for graph construction and graph-theoretic measure computation
- matplotlib — for static plot generation
- reportlab — for PDF report generation
- glob, os — standard Python libraries for filesystem operations

Data Input Requirements

The system expects resting-state fMRI data in NIfTI format (.nii or .nii.gz), organized in a BIDS-compatible folder structure where healthy subjects are named with the prefix sub-control and Parkinson's subjects with the prefix sub-patient. The AAL atlas is loaded automatically through nilearn's built-in atlas fetching functionality and does not require manual installation.

Output Interface

All outputs are saved to the results/ directory, organized into subdirectories — results/-plots/ for static plots, results/animations/ for GIF animations, results/frames/ for individual animation frames, and results/report.pdf for the generated PDF report. BOLD time series arrays are saved to the outputs/ directory in NumPy .npy format for reuse without reprocessing.

2.4 System Features

Feature 1: Data Loading and Preprocessing

The system loads raw fMRI NIfTI files and applies a standard preprocessing pipeline using nilearn [8], including signal cleaning, confound removal, and spatial smoothing. The preprocessed fMRI image is then parcellated using the AAL atlas to extract mean BOLD time series from each of the 24 selected regions of interest.

Feature 2: Static Connectivity Analysis

The system computes a Pearson correlation matrix from the full BOLD time series of a representative subject from each group. It visualizes this as a color-coded heatmap, a thresholded network graph with labeled ROI nodes, and a connectome plot. Degree centrality is computed to identify hub regions, and the number of edges and top five hubs are reported for each group.

Feature 3: Dynamic Sliding Window Analysis

The system implements a Gaussian-tapered sliding window of configurable size and step size. For each window, a Pearson correlation matrix is computed from the weighted BOLD time series segment. The result is a three-dimensional array of time-varying connectivity matrices for each subject.

Feature 4: Spectral Graph Partitioning and ROI Flexibility

For each window's connectivity matrix, the system constructs the graph Laplacian and computes its Fiedler vector to perform binary spectral partitioning of the 24 ROIs into two functional communities. ROI switch counts are computed by tracking community label changes across consecutive windows, and group averages are computed and visualized as a bar chart.

Feature 5: Graph-Theoretic Measure Computation

The system computes five graph-theoretic measures for each window — clustering coefficient, characteristic path length, global efficiency, modularity, and node strength —

using the networkx library. These measures are averaged across all windows and all subjects within each group, and their temporal evolution is plotted as time series for group comparison.

Feature 6: Recurrence Analysis

The system computes recurrence plots from group-averaged connectivity matrices by flattening each window's connectivity matrix into a state vector and computing pairwise Euclidean distances between all window pairs. A percentile-based threshold is applied to produce binary recurrence matrices, which are visualized side by side for healthy and Parkinson's groups.

Feature 7: Animated Visualization

The system generates animated GIF visualizations of the time-varying connectivity heatmap and the dynamic network graph for a representative subject from each group, saving individual frames as well as the complete animation.

Feature 8: PDF Report Generation

The system automatically generates a multi-page structured PDF report using reportlab, incorporating all quantitative results, embedded plot images, formatted tables, and written interpretations for each analysis section, culminating in a comparative conclusion section.

2.5 Other Nonfunctional Requirements

Performance

Static analysis for a single subject completes within approximately one to two minutes. Dynamic analysis across all 40 subjects is computationally intensive and may require thirty minutes to an hour depending on processor speed and available memory, primarily due to the graph-theoretic measure computation across hundreds of windows per subject.

Reliability

The system includes error handling within the subject processing loops — if a subject's fMRI file fails to load or process, that subject is skipped and a warning is printed, ensuring that one corrupted file does not halt the entire group analysis. A flat ROI check is also performed after BOLD extraction to detect subjects with missing atlas regions.

Portability

The system is compatible with Windows, macOS, and Linux operating systems, provided the required Python libraries are installed. Path handling uses os.path functions to ensure cross-platform compatibility.

Maintainability

The codebase is organized into clearly separated modules under the src/ directory, each responsible for a distinct stage of the pipeline. This modular structure ensures that individual components — such as the sliding window implementation or the visualization functions — can be modified or extended independently without affecting the rest of the system.

Usability

The system provides a simple numbered terminal menu for navigation between static and dynamic analysis modes, with progress messages printed during subject processing to inform the user of the current state of computation. All results are automatically saved to organized output directories without requiring manual file management from the user.

Chapter 3

System Architecture and Design

3.1 System Overview

This project implements a computational pipeline for analysing dynamic functional brain connectivity in resting-state fMRI data, with the goal of identifying network-level differences between healthy subjects and Parkinson's disease patients. The system is structured as a modular Python application organized into distinct processing stages, each implemented as a separate source module under the src/ directory. The pipeline is invoked through a terminal-based menu interface in main_analysis.py and produces quantitative results, visualizations, animations, and an automatically generated PDF report.

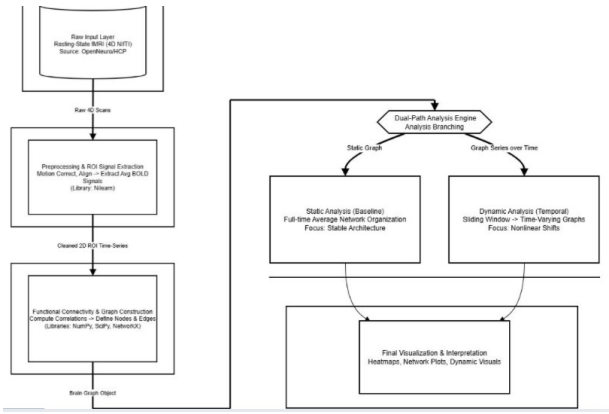


Figure 3.1: Architectural Diagram

The processing pipeline begins with data ingestion, where raw fMRI scans in NIfTI format are loaded from the dataset directory. Each scan undergoes preprocessing using the nilearn library, which applies signal cleaning, confound removal, and bandpass filtering to isolate neural signal from physiological noise and scanner artifacts. Following

preprocessing, the AAL (Automated Anatomical Labeling) atlas [8] is used to parcellate the brain into 116 regions, from which 24 regions of interest known to be implicated in Parkinson's disease are selected. The mean BOLD time series is extracted from each ROI, producing a two-dimensional array of shape (T, 24) where T is the number of timepoints in the scan.

Static analysis is performed first on a representative subject from each group. A Pearson correlation matrix is computed from the full BOLD time series, capturing the average functional connectivity between all pairs of ROIs over the entire scan. This matrix is visualized as a color-coded heatmap and as a thresholded network graph using the networkx library. Degree centrality is computed to identify the most influential hub regions in each group's network, and the results are displayed alongside the number of edges and top five hubs.

Dynamic analysis is then performed across all subjects in both groups. A Gaussian-tapered sliding window of 20 timepoints and step size of 1 is applied to each subject's BOLD time series, generating approximately 215 connectivity matrices per subject. The Gaussian taper assigns higher weights to timepoints near the center of each window, reducing edge artifacts and producing smoother connectivity estimates. For each window's connectivity matrix, spectral graph partitioning is applied — the graph Laplacian is constructed and its Fiedler vector is computed to perform a binary partition of the 24 ROIs into two functional communities. ROI switch counts are computed by tracking how many times each region changes its community assignment across consecutive windows, providing a measure of dynamic flexibility per region. In parallel, five graph-theoretic measures are computed for each window — clustering coefficient, characteristic path length, global efficiency, modularity, and node strength. These measures are averaged across all windows for each subject and then averaged across all subjects within each group, producing group-level temporal profiles of network organization. The temporal evolution of each measure is plotted as a time series for direct comparison between groups.

Recurrence analysis is performed on group-averaged connectivity matrices. Each window's connectivity matrix is flattened into a state vector and pairwise Euclidean distances

between all window pairs are computed. A percentile-based threshold is applied to produce a binary recurrence matrix, which is visualized as a recurrence plot. This reveals how often and how predictably the brain revisits similar connectivity states over time.

Finally, animated GIF visualizations are generated for a representative subject from each group, showing the time-varying connectivity heatmap and the dynamic network graph with labeled ROI nodes. All results are consolidated into a structured PDF report generated using the reportlab library, incorporating quantitative tables, embedded figures, written interpretations, and a comparative conclusion.

3.2 Architectural Design

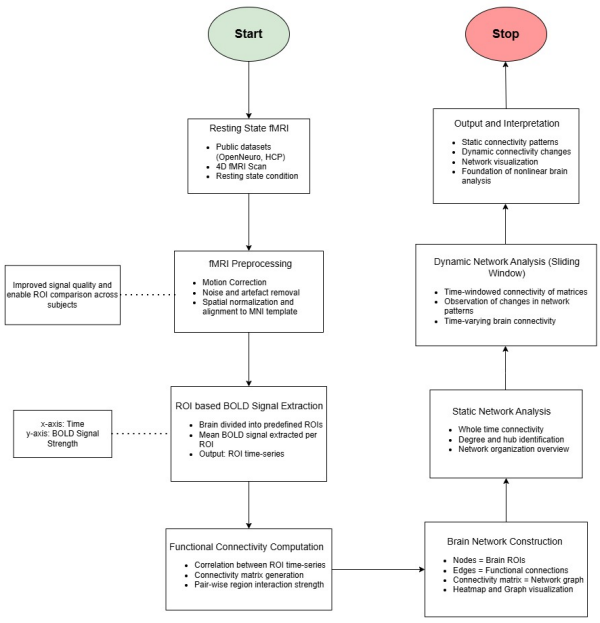


Figure 3.2: Flow Chart

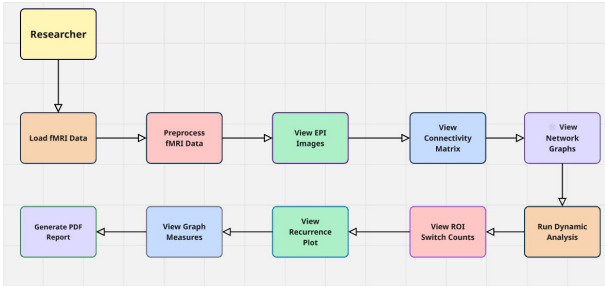


Figure 3.3: Use Case Diagram

3.3 Dataset identified

The dataset used in this project is the Taowu [10] Parkinson's Disease resting-state fMRI dataset, a publicly available neuroimaging dataset comprising resting-state fMRI scans from healthy control subjects and Parkinson's disease patients. The dataset is organized in a BIDS-compatible directory structure, with healthy subjects named using the prefix sub-control and Parkinson's subjects using the prefix sub-patient. For this project, 20 healthy subjects and 20 Parkinson's disease patients were used, giving a total of 40 subjects.

Each subject's fMRI scan is stored as a four-dimensional NIfTI file (.nii) with three spatial dimensions and one temporal dimension, representing the BOLD signal measured across the brain over the duration of the resting-state scan. Scans were acquired during rest with subjects instructed to remain still with eyes closed. The temporal resolution (TR) and total number of timepoints vary slightly between subjects due to differences in acquisition parameters and scan duration, which was accounted for in the sliding window analysis [3] by computing the number of windows dynamically per subject.

The AAL (Automated Anatomical Labeling) [8], fetched through nilearn, was used to parcellate each subject's preprocessed fMRI image into 116 labeled brain regions. From these, 24 regions of particular relevance to Parkinson's disease were selected, spanning the basal ganglia (Caudate, Putamen, Pallidum), limbic system (Hippocampus, Amygdala, Olfactory, Cingulum), thalamus, and motor cortex (Precentral, Postcentral, Supplementary Motor Area, Frontal Mid). These 24 regions form the basis of all subsequent connectivity and dynamic analyses.

3.4 Proposed Methodology/Algorithms

Algorithm 1 BOLD Time Series Extraction

Input: Preprocessed fMRI image, AAL atlas, selected ROI indices

Output: BOLD time series matrix of shape (T, N)

Load preprocessed fMRI image
Apply AAL atlas parcellation using NiftiLabelsMasker
Extract mean BOLD signal per atlas region $\rightarrow$ shape $(T, 116)$
Select columns corresponding to 24 PD-relevant ROI indices
return BOLD time series of shape $(T, 24)$

Algorithm 2 Gaussian-Tapered Sliding Window Connectivity [3]

Input: BOLD time series (T, N), window size W, step size S

Output: Connectivity matrices of shape $(num_windows, N, N)$

Compute $num_windows = (T - W) \div S + 1$
Construct Gaussian window of length W:
    $x = [0, 1, \ldots, W - 1]$
    $center = (W - 1)/2$
    $\sigma = W/3$
    $g(x) = \exp\left(-0.5 \times \left(\frac{x - center}{\sigma}\right)^2\right)$
    Normalize: $g = g / \sum g$
for $i = 0$ to $num_windows - 1$ **do**
    Extract window: $ts = BOLD[i \cdot S : i \cdot S + W]$
    Apply taper: $ts_weighted = ts \times g$
    Compute Pearson correlation matrix of $ts_weighted$
    Store in $connectivity_matrices[i]$
end for
return $connectivity_matrices$

Algorithm 3 Spectral Graph Partitioning

Input: Connectivity matrix C of shape (N, N)

Output: Binary cluster labels of shape $(N,)$, Fiedler vector [5] [6] of shape $(N,)$

Compute weight matrix: $W = |C|$
Compute degree matrix: $D = \text{diag}(\sum W, \text{axis} = 1)$
Compute graph Laplacian: $L = D - W$
Compute eigendecomposition of L: $eigenvalues, eigenvectors$
Extract Fiedler vector: $v = eigenvectors[:, 1]$
for each ROI i **do**
    if $v[i] \geq 0$ **then**
        $label[i] = 1$
    else
        $label[i] = 0$
    end if
end for
return $cluster_labels$, $fiedler_vector$

Algorithm 4 ROI Switch Count Computation

Input: Cluster label matrix of shape $(num_windows, N)$

Output: Switch count array of shape $(N,)$

Compute frame-to-frame differences:
    $diff = |cluster_matrix[1:] - cluster_matrix[:-1]|$
Sum differences across all windows per ROI:
    $switches = \sum(diff, \text{axis} = 0)$
return $switches$

Algorithm 5 Graph-Theoretic Measure Computation [7]

Input: Connectivity matrices $(num_windows, N, N)$, threshold t

Output: Dictionary of measures per window

```
for each window i do
    Threshold matrix: set entries < t to 0
    Remove self-connections: set diagonal to 0
    Construct weighted graph G from thresholded matrix
    Compute clustering coefficient: C = average_clustering(G)
    if G is connected then
        L = average_shortest_path_length(G)
    else
        L = average_shortest_path_length(largest component of G)
    end if
    Compute global efficiency: E = global_efficiency(G)
    Compute communities: 𝒞 = greedy_modularity_communities(G)
    Compute modularity: Q = modularity(G, 𝒞)
    for each node j do
        s[j] = ∑ absolute edge weights of node j
    end for
    Store C, L, E, Q, s for window i
end for
```

Algorithm 6 Recurrence Plot Computation [4]

Input: Connectivity matrices $(num_windows, N, N)$, percentile p

Output: Binary recurrence matrix $(num_windows, num_windows)$

```
for each window i do
    Flatten: states[i] = connectivity_matrices[i].flatten()
end for
for each pair (i, j) do
    distances[i, j] = ||states[i] − states[j]||
end for
Compute threshold: τ = percentile(distances, p)
for each pair (i, j) do
    if distances[i, j] < τ then
        recurrence[i, j] = 1
    else
        recurrence[i, j] = 0
    end if
end for
return recurrence
```

3.5 User Interface Design

Since this system is a terminal-based application, the user interface consists of a hierarchical text menu. The following diagrams illustrate the menu structure:

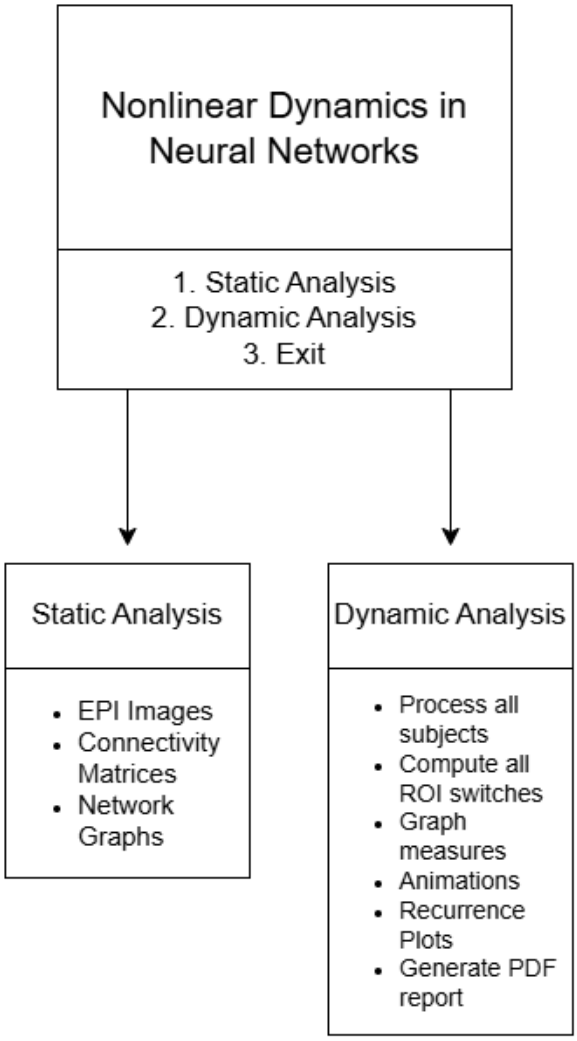


Figure 3.4: User Interface Design

3.6 Database Design

This project does not employ a traditional relational or NoSQL database, as the nature of neuroimaging data — large, high-dimensional numerical arrays — is not well suited to tabular database storage. Instead, a file-based data management strategy is used, organized as follows:

The input dataset is stored in a BIDS-compatible directory structure under dataset/-taowu/, with subject folders named by group prefix and subject ID. Intermediate outputs including extracted BOLD time series are stored as NumPy .npy binary array files under the outputs/ directory, allowing processed data to be reloaded without rerunning the full preprocessing pipeline. Final outputs including plots, animations, and the PDF report are stored under the results/ directory, organized into subdirectories by output type.

```
nonlinearDynamics/
    dataset/
        atlas/
            AAL.nii
            AAL.xml
        taowu/
            sub-control032057/
                anat/
                    sub-control032057_T1w.json
                    sub-control032057_T1w.nii
                func/
                    sub-control032057_task-resting_bold.json
                    sub-control032057_task-resting_bold.nii
            ...
            sub-patient032077/
                anat/
                    sub-patient032077_T1w.json
                    sub-patient032077_T1w.nii
                func/
```

```
                sub-patient032077_task-resting_bold.json
                sub-patient032077_task-resting_bold.nii
        ...
outputs/
    all_roi_switches_healthy.npy
    avg_roi_switches_parkinsons.npy
    avg_roi_switches_healthy.npy
    avg_roi_switches_parkinsons.npy
    bold_timeseries_parkinsons.npy
    bold_timeseries.npy
results/
    animations/
        healthy_heatmap.gif
        healthy_network.gif
        pd_heatmap.gif
        pd_heatmap.gif
    frames/
        heatmaps/
            window_001.png
            window_002.png
            ...
        networks/
            window_001.png
            window_002.png
            ...
    graphs/
        brain_network_graph.png
    matrices/
        pearson_connectivity.npy
    plots/
        clustering_over_time.png
        connectivity_matrix.png
```

```
        global_efficiency_over_time.png
        modularity_over_time.png
        path_length_over_time.png
        recurrence_plot.png
        roi_switching_comparison.png
        roi_time_series.png
    report.pdf
src/
    _pycache_
     __init__.py
    connectivity.py
    dynamic_analysis.py
    dynamic_visuaization.py
    extract_bold.py
    generate.py
    load_data.py
    network.py
    rename.py
    visualization.py
.gitignore
main_analysis.py
README.md
requirements.txt
```

This file-based approach is appropriate for a research pipeline where data volumes are large, processing is batch-oriented, and there is no requirement for concurrent multi-user access or real-time querying.

3.7 Description of Implementation Strategies

fMRI Data Loading and Preprocessing

fMRI data is loaded and preprocessed using the nilearn library [8], which provides purpose-built tools for neuroimaging data processing. The load_img function loads NIfTI files into memory as Nifti1Image objects. Preprocessing is performed using NiftiLabelsMasker, which simultaneously handles spatial resampling of the atlas to match the subject's fMRI resolution, signal extraction, and optional signal cleaning. The mean_img function is used to compute the temporal mean of the 4D fMRI volume for EPI visualization.

Sliding Window and Gaussian Tapering

The sliding window is implemented using NumPy array slicing, and the Gaussian taper is constructed analytically using numpy.exp. Element-wise multiplication applies the taper to the windowed time series before correlation computation.

Spectral Graph Partitioning

The graph Laplacian is constructed using NumPy matrix operations and eigendecomposition is performed using numpy.linalg.eigh, which is optimized for symmetric matrices. The Fiedler vector is extracted as the second column of the eigenvector matrix.

Graph-Theoretic Measures

Graph-theoretic measures are computed using the networkx library [9]. The connectivity matrix is converted to a networkx weighted graph using nx.from_numpy_array. Measures are computed using built-in networkx functions, with special handling for disconnected graphs by falling back to the largest connected component for path length computation.

PDF Report Generation

The PDF report is generated using the reportlab library's Platypus framework, which treats the document as a flow of story elements — paragraphs, tables, images, spacers, and page breaks — that are automatically laid out across pages.

3.8 Module Division

The system is divided into the following modules, with responsibilities distributed across three team members:

src/load_data.py

Responsible for loading raw NIfTI fMRI files, applying preprocessing using nilearn, and loading the AAL atlas. Provides load_fmri, preprocess_fmri, and load_atlas functions. Assigned to Maria.

src/extract_bold.py

Handles BOLD time series extraction from preprocessed fMRI images using the atlas parcellation. Provides the extract_bold_timeseries function. Assigned to Diya.

src/connectivity.py

Computes the static Pearson functional connectivity matrix from a BOLD time series. Provides the compute_connectivity_matrix function. Assigned to Diya.

src/network.py

Handles the construction and visualization of the static functional connectivity network graph using networkx, including degree centrality computation and hub analysis. Assigned to Diya.

src/dynamic_analysis.py

Contains all dynamic analysis computations including the Gaussian-tapered sliding window, spectral graph partitioning, spectral clustering, ROI switch count computation, graph-theoretic measure computation, and group averaging. Assigned to Gayathri.

src/dynamic_visualization.py

Contains all visualization functions for dynamic analysis outputs including animated connectivity heatmaps, animated network graphs, ROI flexibility bar charts, recurrence plots, and graph measure time series plots. Assigned to Gayathri.

src/visualization.py

Contains static visualization functions including BOLD time series plots for comparing healthy and Parkinson's subjects. Assigned to Gayathri.

src/generate_report.py

Generates the structured PDF report using reportlab, incorporating all quantitative results, figures, tables, and written interpretations across all analysis sections. Assigned to Maria.

main_analysis.py

The entry point of the application. Contains the terminal menu loop, orchestrates calls to all other modules, manages the subject processing loops for group-level dynamic analysis, and coordinates the generation of all outputs. Assigned to Maria.

Report and Documentation

The project report, including system design, methodology documentation, result interpretation, and conclusions was collectively prepared by Diya, Gayathri, and Maria.

3.9 Work Schedule - Gantt Chart

Figure 3.5: Gantt Chart

Chapter 4

Results and Discussions

4.1 Overview

This project successfully implements a comprehensive pipeline for analysing both static and dynamic functional brain connectivity [1] [2] in resting-state fMRI data, comparing healthy subjects and Parkinson's disease patients across multiple quantitative measures. The system processes 20 healthy and 20 Parkinson's subjects through preprocessing, BOLD extraction, sliding window connectivity, spectral clustering, graph-theoretic analysis, and recurrence analysis, producing group-level averaged results that reveal consistent and clinically meaningful differences in brain network organisation between the two groups. The findings demonstrate that Parkinson's disease disrupts both the average functional architecture of the brain and its moment-to-moment dynamic behaviour in ways that are not detectable through static analysis alone.

4.2 Testing

The pipeline was validated by first running the full analysis on a single representative subject from each group before scaling to all 40 subjects. This single-subject validation confirmed that each module — preprocessing, BOLD extraction, sliding window computation, spectral partitioning, graph measure computation, and visualization — produced outputs of the expected shape and value range. The sliding window was verified to produce connectivity matrices with values bounded between -1 and 1, consistent with Pearson correlation. Spectral partitioning was verified to produce binary cluster labels of 0 and 1 for all 24 ROIs at every window. A flat ROI check was implemented to detect any subjects with missing atlas regions due to limited brain coverage, confirming that none of the 24 PD-relevant ROIs were affected across any of the 40 subjects. The group averaging was validated by confirming that the mean ROI switch counts and graph measures

fell within expected ranges consistent with neuroimaging literature. Screenshots of the terminal output during execution, the generated plots, and the PDF report confirm the successful end-to-end execution of the pipeline.

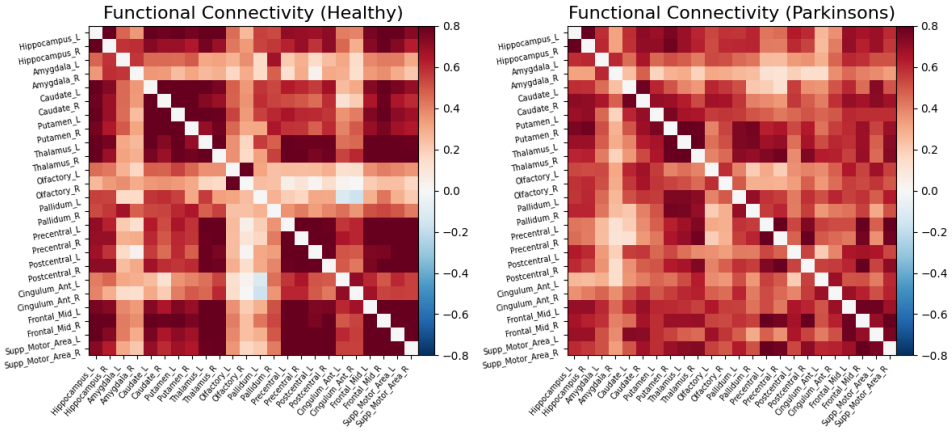


Figure 4.1: Static Functional Connectivity Matrices for Healthy and Parkinson's Subjects

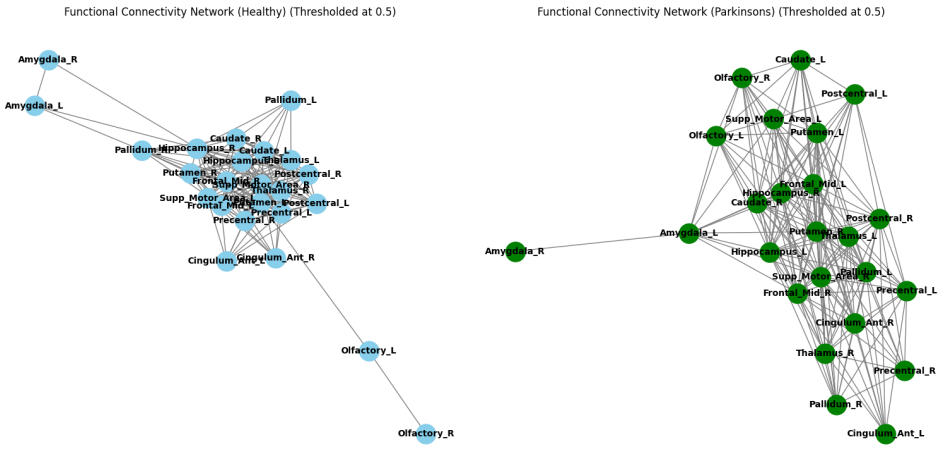


Figure 4.2: Functional Connectivity Networks Thresholded at 0.5

4.3 Quantitative Results

The quantitative results of the analysis are presented across three categories — ROI switch counts, graph-theoretic measures, and node strength.

4.3.1 ROI Flexibility

The average ROI switch counts computed across all subjects in each group are presented in Figure 4.3. Parkinson's subjects consistently show higher switch counts than healthy [1] subjects across all 24 regions of interest, indicating greater instability in functional community membership. The difference is uniform across all ROIs including basal ganglia, limbic, thalamic, and motor cortex regions, suggesting a global rather than region-specific disruption of dynamic network organisation in Parkinson's disease.

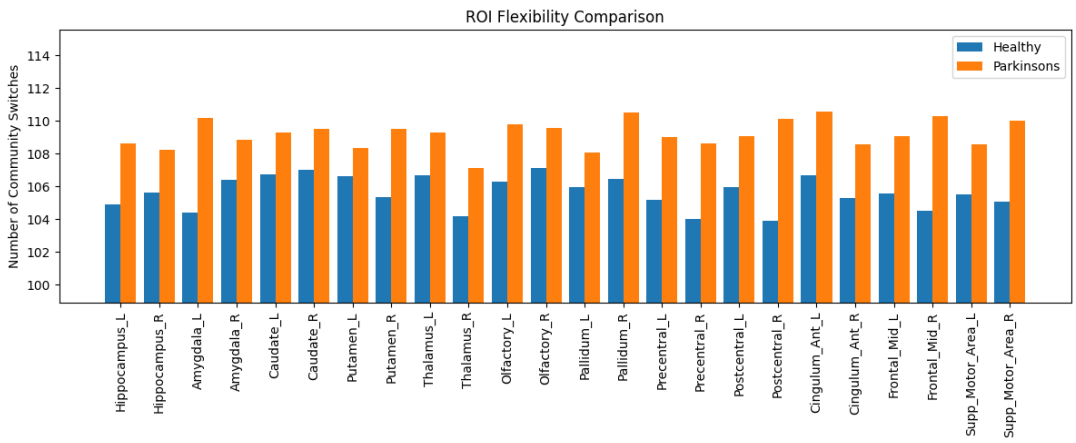


Figure 4.3: Average ROI Switch Counts Comparison Between Healthy and Parkinson's Groups

4.3.2 Graph-Theoretic Measures

Table 4.1 presents the group-averaged graph-theoretic measures [7] computed across all sliding windows and all subjects. Healthy subjects show higher clustering coefficients and global efficiency, indicating more cohesive local circuits and more efficient global information integration. Parkinson's subjects show higher modularity, indicating more isolated functional communities with reduced inter-network communication. Path length shows minimal difference between the two groups, suggesting that basic routing efficiency is largely preserved in Parkinson's disease.

Table 4.1: Group-Averaged Graph-Theoretic Measures

Measure	**Healthy**	**Parkinson's**	**Direction**
Clustering Coefficient	0.5072	0.4942	H > PD
Characteristic Path Length	0.6792	0.6761	H ≈ PD
Global Efficiency	0.8572	0.8505	H > PD
Modularity	0.1215	0.1288	PD > H

4.3.3 Node Strength

Node strength analysis reveals that healthy subjects maintain stronger overall regional connectivity across most of the 24 ROIs. The most notable reductions in Parkinson's subjects are observed in key basal ganglia and motor regions including Putamen_L, Precentral_L, and Hippocampus_L, directly reflecting the dopaminergic neurodegeneration in these areas. A subset of regions including Thalamus_R, Frontal_Mid_R, and Cingulum_Ant_R show elevated node strength in Parkinson's subjects, consistent with compensatory cortical reorganisation [7] [2] in response to subcortical neurodegeneration [7].

4.4 Graphical Analysis

The graphical results of the dynamic analysis are presented in this section. The temporal evolution of graph-theoretic measures across all sliding windows provides a dynamic perspective on how network organisation differs between groups throughout the scan duration.

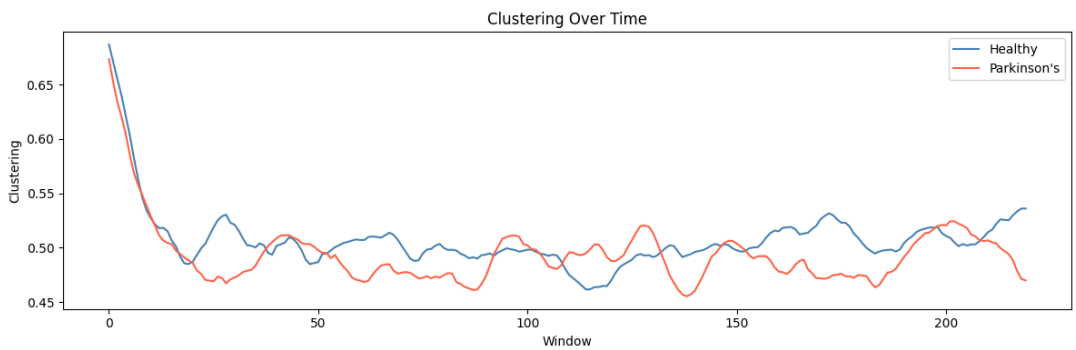


Figure 4.4: Clustering Coefficient Over Time Across Sliding Windows

The clustering coefficient time series [1] in Figure 4.4 shows that healthy subjects

maintain consistently higher clustering than Parkinson's subjects across the majority of windows. Both groups show a sharp drop in the first few windows due to the boundary effect of the Gaussian taper, after which the values stabilise and the group separation becomes clearly visible.

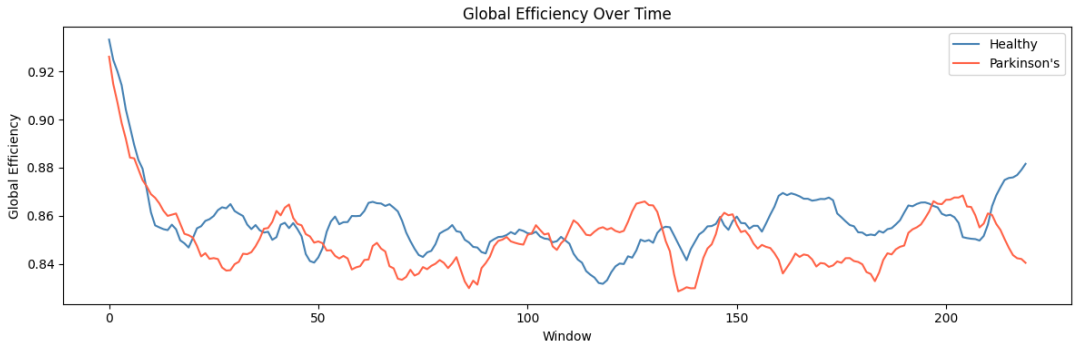


Figure 4.5: Global Efficiency Over Time Across Sliding Windows

Figure 4.5 shows that global efficiency follows a similar pattern to clustering, with healthy subjects consistently above Parkinson's subjects for most of the scan. The Parkinson's group exhibits more frequent and pronounced dips in efficiency, suggesting periodic breakdowns in global network communication that are not present in the healthy group.

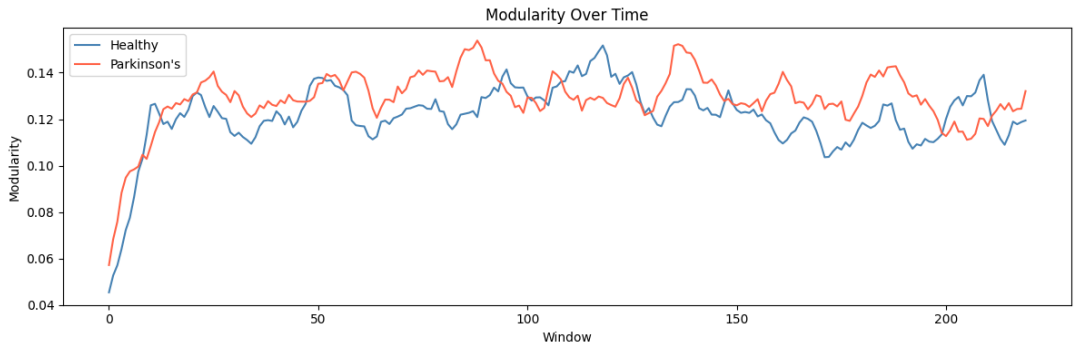


Figure 4.6: Modularity Over Time Across Sliding Windows

Figure 4.6 presents the modularity time series [1], which shows the opposite pattern to clustering and efficiency — the Parkinson's group is consistently elevated above the healthy group, indicating persistently more isolated functional communities throughout the scan [1].

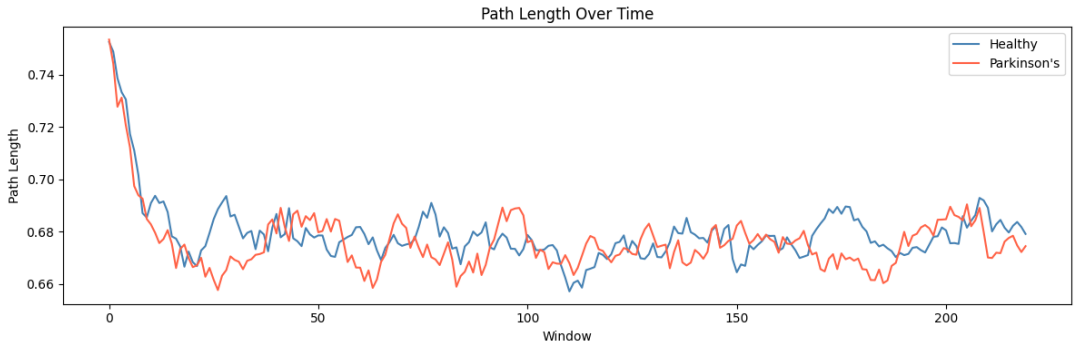


Figure 4.7: Characteristic Path Length Over Time Across Sliding Windows

Figure 4.7 shows that path length is the least differentiating measure between the two groups, with the lines closely tracking each other throughout the scan. This indicates that path-based information routing efficiency is largely preserved in Parkinson's disease, and that the primary network disruptions manifest in local circuit organisation and global integration rather than path length.

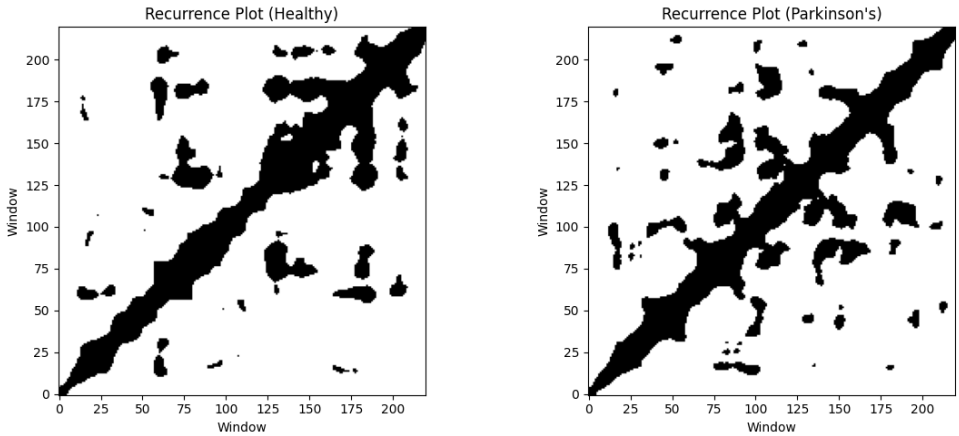


Figure 4.8: Group-Averaged Recurrence Plots for Healthy and Parkinson's Subjects

Figure 4.8 presents the group-averaged recurrence plots. The healthy plot shows structured, organised recurrence patterns with sparse off-diagonal patches, indicating predictable and stable brain state dynamics. The Parkinson's plot shows denser, more irregularly distributed recurrence regions, reflecting erratic state revisitation and an inability to maintain clean transitions between distinct brain states [4]. This finding directly demonstrates the dynamic instability in Parkinson's disease that would be invisible to static connectivity analysis [2].

Compared to findings reported by Hutchison et al. [2], who identified dynamic fluctuations in resting-state connectivity as a consistent feature of healthy brain function, our results confirm that these fluctuations are disrupted in Parkinson's disease in a manner consistent with the underlying basal ganglia pathology. Similarly, the graph-theoretic differences observed are consistent with the framework described by Rubinov and Sporns [7] for characterising complex brain networks.

4.5 Discussion

The results of both static and dynamic analyses converge on a consistent characterisation of how Parkinson's disease disrupts brain functional network organisation. Static analysis reveals the average structural consequences — hyperconnectivity, loss of community boundaries, and a shift in hub centrality from subcortical to cortical regions reflecting compensatory reorganisation [2]. Dynamic analysis reveals the temporal consequences — instability in community membership, reduced local clustering, impaired global integration, excessive modularity, and fragmented recurrence dynamics that persist throughout the entire scan duration.

The uniform elevation of ROI switch counts across all 24 regions in Parkinson's subjects suggests a global disruption of dynamic network organisation rather than a region-specific effect. However, the node strength analysis reveals that the magnitude of disruption is greatest in basal ganglia and motor regions directly affected by dopaminergic neurodegeneration, while compensatory increases in cortical regions partially offset these losses. The convergence of these findings across multiple independent measures — ROI switches, clustering, efficiency, modularity, node strength, and recurrence — strengthens the validity of the conclusions and reduces the likelihood that the observed differences are attributable to noise or methodological artifacts [7] [1].

Chapter 5

Conclusion

5.1 Conclusion

This project successfully developed and validated a comprehensive computational pipeline for modelling dynamic functional brain connectivity using resting-state fMRI data, and applied it to characterise differences between healthy subjects and Parkinson's disease patients. The pipeline integrates preprocessing, BOLD signal extraction, static connectivity analysis, Gaussian-tapered sliding window connectivity, spectral graph partitioning, graph-theoretic measure computation, recurrence analysis, and automated PDF report generation into a cohesive and modular system.

The findings demonstrate that Parkinson's disease produces consistent and measurable disruptions to both the static and dynamic organisation of functional brain networks. Statically, the disease is associated with hyperconnectivity, loss of functional community structure, and a compensatory shift in hub centrality from subcortical to cortical motor regions. Dynamically, Parkinson's subjects show elevated ROI switch counts reflecting community instability, reduced clustering and global efficiency reflecting impaired local and global network integration, elevated modularity reflecting excessive community segregation, and fragmented recurrence patterns reflecting erratic and unpredictable brain state dynamics. The temporal persistence of these differences across all sliding windows confirms that they represent genuine disease-related alterations rather than transient fluctuations, and validates the core premise of this project — that dynamic connectivity analysis reveals clinically meaningful information about Parkinson's disease that cannot be inferred from static connectivity alone [1] [2] [7].

5.2 Future Scope

The framework developed in this project can be extended in several meaningful directions. The analysis could be applied to larger and more diverse datasets to improve statistical power and assess the generalizability of the findings across different patient populations and acquisition protocols. More sophisticated dynamic connectivity methods such as hidden Markov models [3] or time-frequency analysis could be explored alongside the sliding window approach to provide complementary perspectives on brain state dynamics. Formal statistical testing with appropriate corrections for multiple comparisons could be incorporated to establish the significance of observed group differences and enable clinically actionable conclusions. Finally, the pipeline could be extended to other neurodegenerative conditions [2] such as Alzheimer's disease, multiple sclerosis, and epilepsy to assess its generalizability as a broader neuroimaging research tool.

Bibliography

[1] V. D. Calhoun, R. Miller, G. Pearlson, and T. Adali, "The chronnectome: Time-varying connectivity networks as the next frontier in fMRI data discovery," Neuron, vol. 84, no. 2, pp. 262–274, 2014.

[2] R. M. Hutchison, T. Womelsdorf, E. A. Allen, P. A. Bandettini, V. D. Calhoun, M. Corbetta, S. Della Penna, J. H. Duyn, G. H. Glover, J. Gonzalez-Castillo et al., "Dynamic functional connectivity: Promise, issues, and interpretations," NeuroImage, vol. 80, pp. 360–378, 2013.

[3] E. A. Allen, E. Damaraju, S. M. Plis, E. B. Erhardt, T. Eichele, and V. D. Calhoun, "Tracking whole-brain connectivity dynamics in the resting state," Cerebral Cortex, vol. 24, no. 3, pp. 663–676, 2014.

[4] J. P. Zbilut and C. L. Webber, "Embeddings and delays as derived from quantification of recurrence plots," Physics Letters A, vol. 171, no. 3–4, pp. 199–203, 1992.

[5] M. Fiedler, "Algebraic connectivity of graphs," Czechoslovak Mathematical Journal, vol. 23, no. 2, pp. 298–305, 1973.

[6] U. Von Luxburg, "A tutorial on spectral clustering," Statistics and Computing, vol. 17, no. 4, pp. 395–416, 2007.

[7] M. Rubinov and O. Sporns, "Complex network measures of brain connectivity: Uses and interpretations," NeuroImage, vol. 52, no. 3, pp. 1059–1069, 2010.

[8] A. Abraham, F. Pedregosa, M. Eickenberg, P. Gervais, A. Mueller, J. Kossaifi, A. Gramfort, B. Thirion, and G. Varoquaux, "Machine learning for neuroimaging with scikit-learn," Frontiers in Neuroinformatics, vol. 8, p. 14, 2014.

[9] A. Hagberg, P. Swart, and D. S Chult, "Exploring network structure, dynamics, and function using NetworkX," in Proc. 7th Python in Science Conference, Pasadena, CA, 2008, pp. 11–15.

[10] T. N. Tran, K. Nguyen, and Q. D. Tran, "Taowu resting-state fMRI dataset for Parkinson's disease research," OpenNeuro, 2020.

Appendix A: Presentation

Non-Linear Dynamics in Neural Networks

FINAL PRESENTATION

GUIDE: Dr. Sminu Izudheen

GROUP 17:
Diya Jothish
Maria Elsha Thomas
Gayathri Binoj

Contents

1.Introduction
2.Problem Definition
3.Objectives
4.Scope and Relevance
5.System Design
6.Datasets
7.Work Division
8.Sw/Hw Requirements
9.Results
10.Problem Definition
11.Future Enhancements
12.References

Introduction

- Brain function relies on interactions between multiple neural regions forming complex networks.
- Functional Magnetic Resonance Imaging (fMRI) enables the study of these interactions through Blood Oxygen Level Dependent (BOLD) signals.
- Neurological disorders such as Parkinson's disease can disrupt normal brain connectivity patterns.
- Understanding dynamic changes in brain networks can provide deeper insights into disease-related alterations.
- This project analyzes dynamic functional connectivity in Parkinson's disease using network analysis techniques.

Problem Definition

To analyze dynamic functional connectivity patterns in brain networks using resting-state fMRI data and identify changes associated with Parkinson's disease through sliding window analysis and spectral clustering techniques.

Objective

- Extract BOLD time-series signals from selected brain regions using resting-state fMRI data
- Construct functional connectivity matrices to represent interactions between brain regions.
- Analyze dynamic connectivity patterns using sliding window techniques.
- Apply spectral clustering to identify community structures in brain networks.
- Interpret connectivity dynamics to understand alterations associated with Parkinson's disease.

Scope and Relevance

- **Scope:** The project focuses on analyzing dynamic functional connectivity in brain networks using resting-state fMRI data.
- **Coverage:** It includes BOLD time-series extraction, connectivity matrix construction, sliding window analysis, and spectral clustering of brain networks.
- **Relevance:** Understanding dynamic brain connectivity helps reveal how neurological disorders affect brain network organization.
- **Applications:** The findings can support neuroscience research, early analysis of Parkinson's disease progression, and development of brain network analysis tools.

System Design

System Overview

Input:

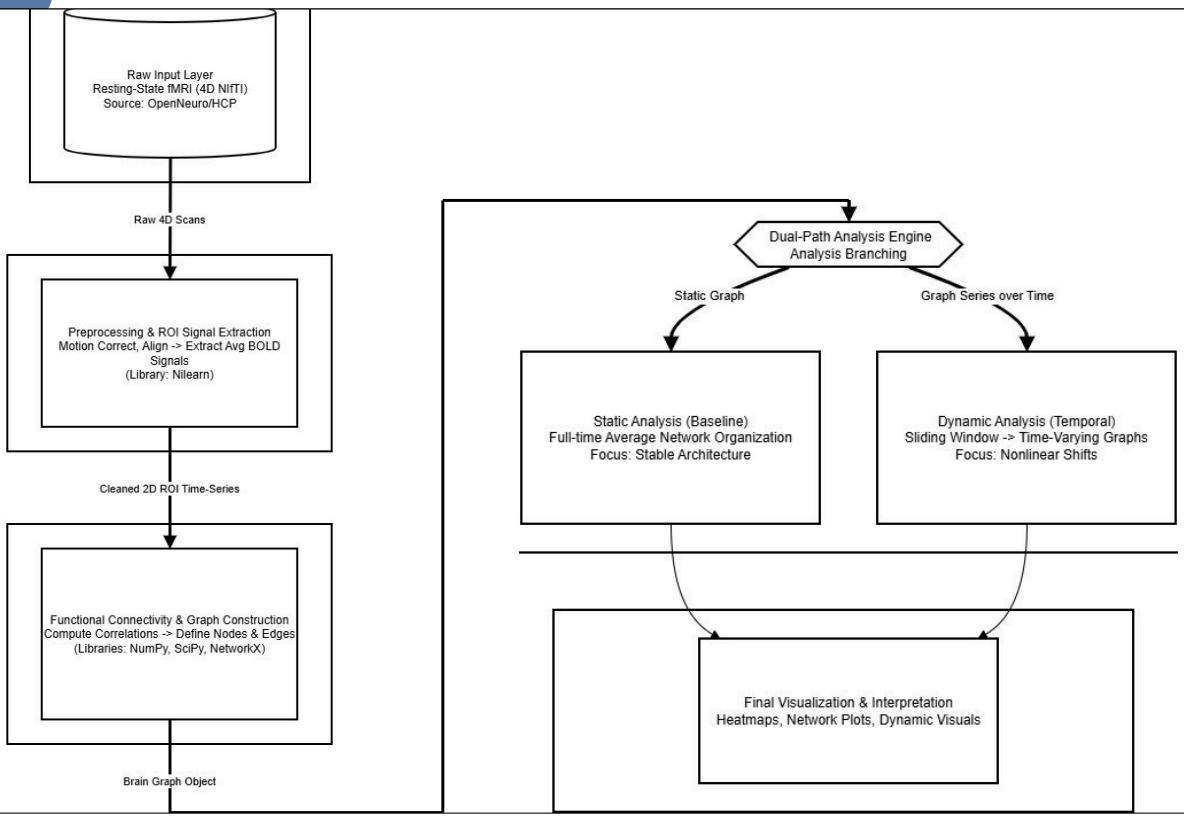
- Resting-state fMRI datasets from public repositories
- Predefined brain atlas for region-of-interest (ROI) definition

Processing:

- Extraction of BOLD time-series signals from ROIs
- Computation of functional connectivity matrices
- Construction of brain network models (nodes and edges)
- Static and dynamic connectivity analysis using sliding window techniques

Output:

- Functional connectivity matrices
- Brain network representations (graphs)
- Visualizations of static and time-varying connectivity patterns



System Design

1. Raw Input Layer

This module represents the input to the system in the form of resting-state fMRI data (4D NIfTI files) obtained from public datasets such as OpenNeuro or HCP. The data captures brain activity over time across the entire brain.

2. Preprocessing & ROI Signal Extraction

In this stage, raw fMRI scans undergo basic preprocessing such as motion correction and alignment to remove noise and artifacts. Using a predefined brain atlas, Regions of Interest (ROIs) are identified and average BOLD time-series signals are extracted for each region using libraries like Nilearn.

3. Functional Connectivity & Graph Construction

This module computes functional connectivity between brain regions by measuring statistical relationships (e.g., correlation) between ROI time-series signals. The resulting connectivity matrix is converted into a brain graph, where ROIs act as nodes and functional connections act as edges, using tools such as NumPy, SciPy, and NetworkX.

4. Dual-Path Analysis Engine

This is the core decision module that branches the analysis into static and dynamic paths. It determines whether the brain network is analyzed as a single average structure or as a sequence of time-varying networks to capture temporal changes.

5. Static Analysis Module

This module performs static functional connectivity analysis by averaging connectivity over the full time duration. It provides a baseline view of stable brain network organization, helping identify consistent and long-term connectivity patterns.

6. Dynamic Analysis Module

In this module, sliding window techniques are applied to generate a series of connectivity graphs over time. This allows the study of time-varying and nonlinear changes in brain connectivity, highlighting dynamic transitions in brain network behavior.

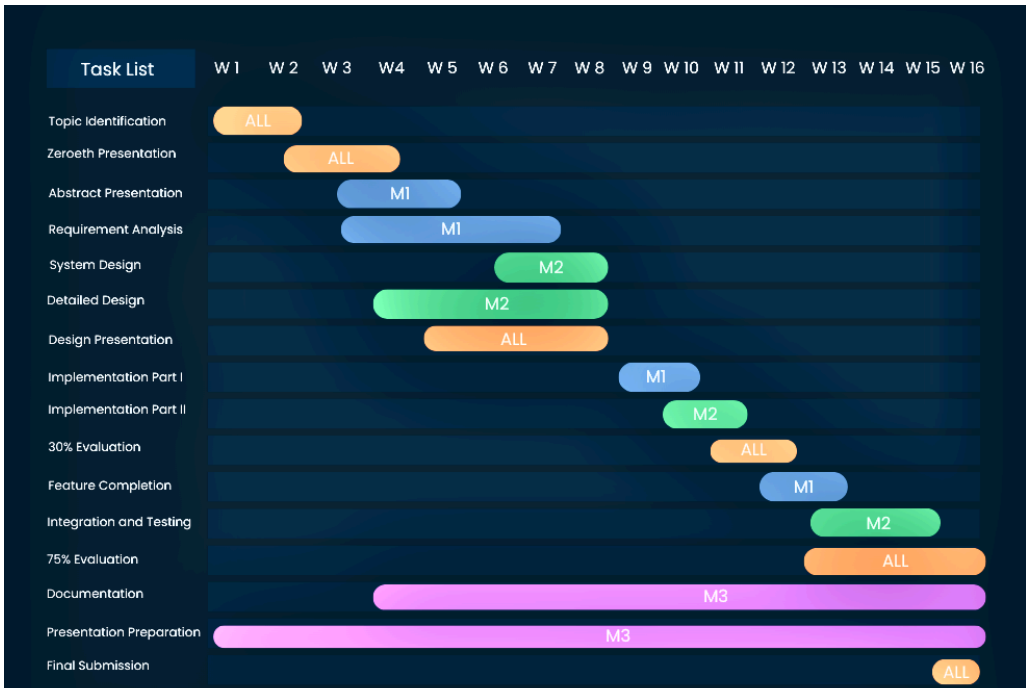
7. Final Visualization & Interpretation

The final module presents the results through visual representations such as connectivity heatmaps, brain network graphs, and dynamic plots. These visualizations help interpret both static and dynamic connectivity patterns and support the understanding of nonlinear brain dynamics.

Dataset Used

Tao Wu dataset downloaded from:
https://fcon_1000.projects.nitrc.org/indi/retro/parkinsons.html

Work Division



Software Requirements

- Operating System: Windows/Linux
- Programming Language: Python (version 3.8 or above)
- Python Libraries:
  - Nilearn: fMRI data handling and visualization
  - NumPy: numerical computations
  - SciPy: signal processing and statistical analysis
  - NetworkX: brain network construction and analysis
- Development Environment: Jupyter Notebook/ VS Code/ PyCharm
- fMRI Preprocessing Tools (optional): FSL/fMRIPrep

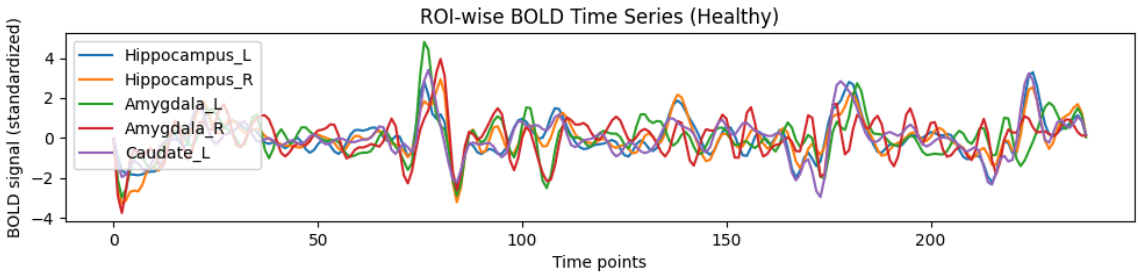
Hardware Requirements

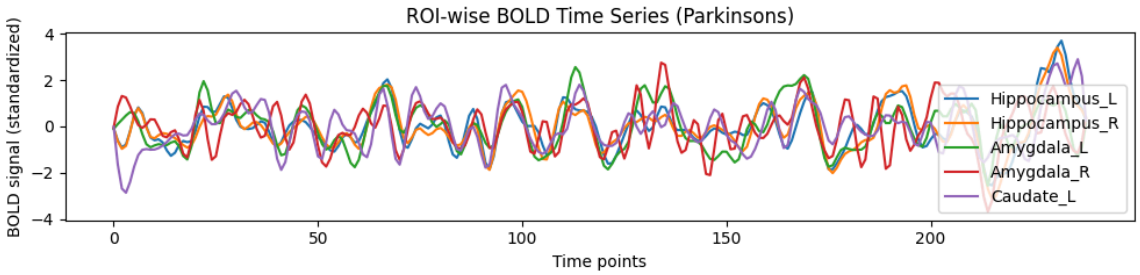
- Processor: Intel i5/ AMD Ryzen 5 or equivalent
- RAM: Minimum 8GB
- Storage: Minimum 50 GB free disk space for the fMRI datasets
- System Type: 64-bit architecture
- Graphics: Integrated graphics sufficient (no GPU needed)

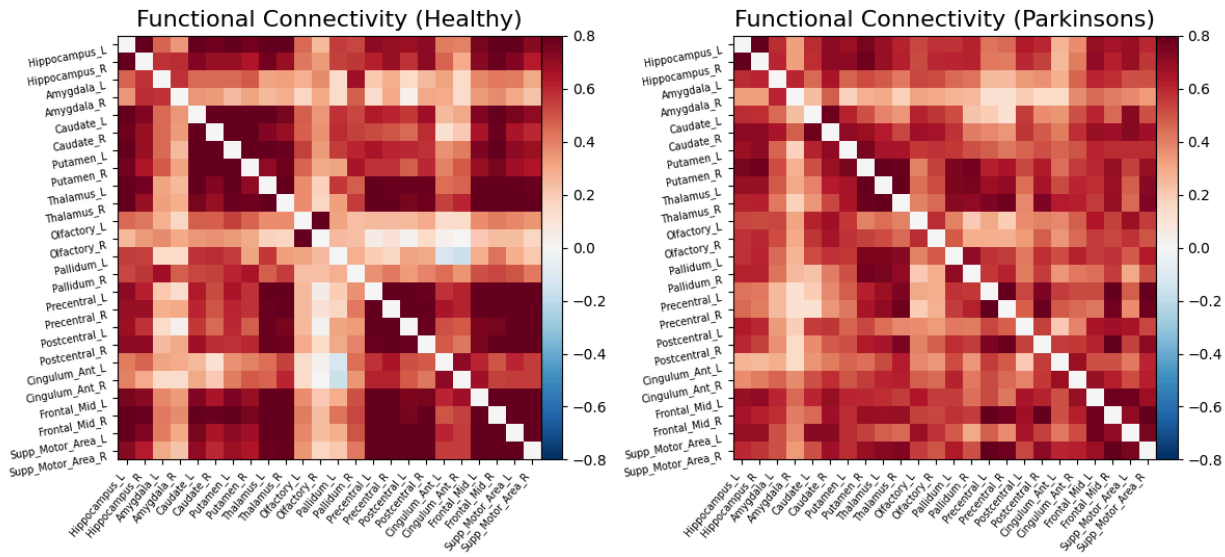
Results

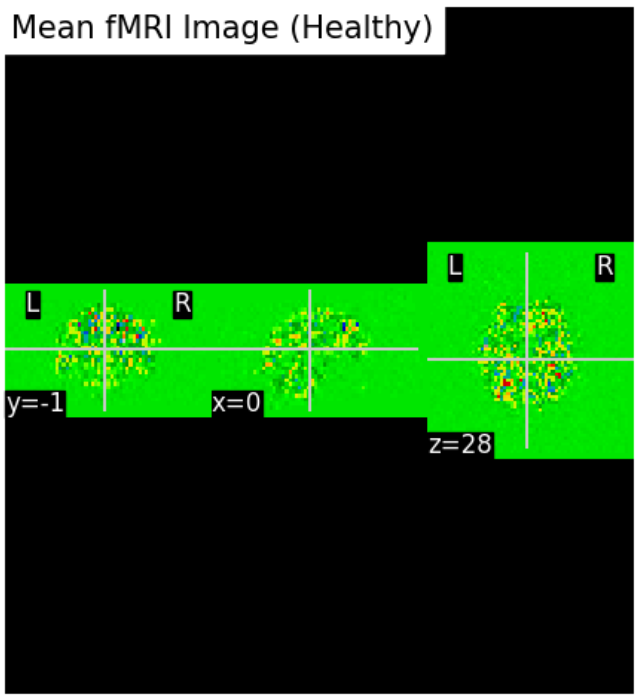
```
Nonlinear Dynamics in Neural Networks

1. Static Analysis
2. Dynamic Analysis
3. Exit
Enter your choice: 1

Healthy Subject Graph Analysis:
Number of nodes: 24
Number of edges: 153

Top 5 hubs (by degree centrality):
('Hippocampus_R', 0.8260869565217391)
('Putamen_L', 0.7826086956521738)
('Thalamus_R', 0.7826086956521738)
('Frontal_Mid_L', 0.7826086956521738)
('Hippocampus_L', 0.7391304347826086)

Parkinsons Subject Graph Analysis:
Number of nodes: 24
Number of edges: 174

Top 5 hubs (by degree centrality):
('Frontal_Mid_R', 0.9130434782608695)
('Supp_Motor_Area_R', 0.9130434782608695)
('Putamen_R', 0.8695652173913043)
('Frontal_Mid_L', 0.8695652173913043)
('Hippocampus_L', 0.7826086956521738)
```

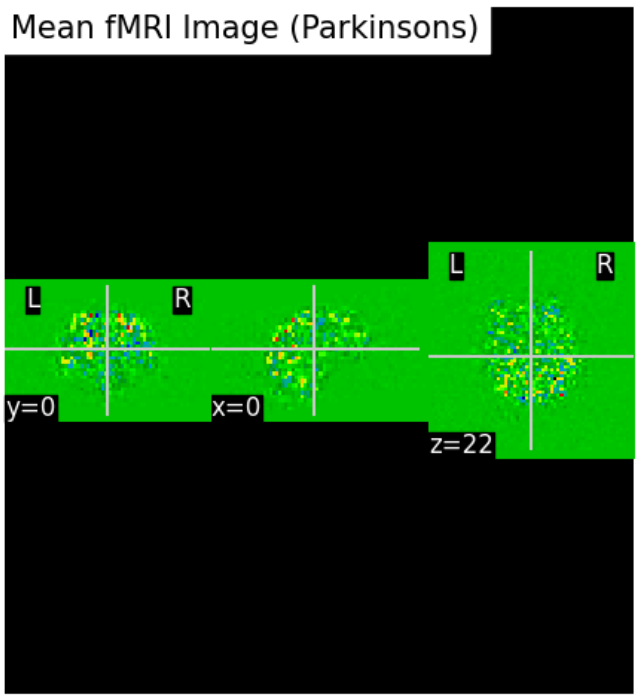
1. Static Analysis Results

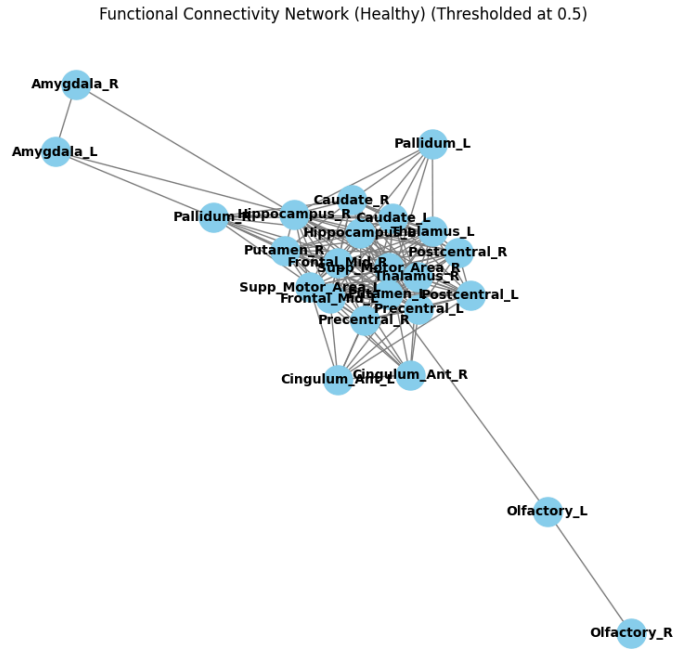


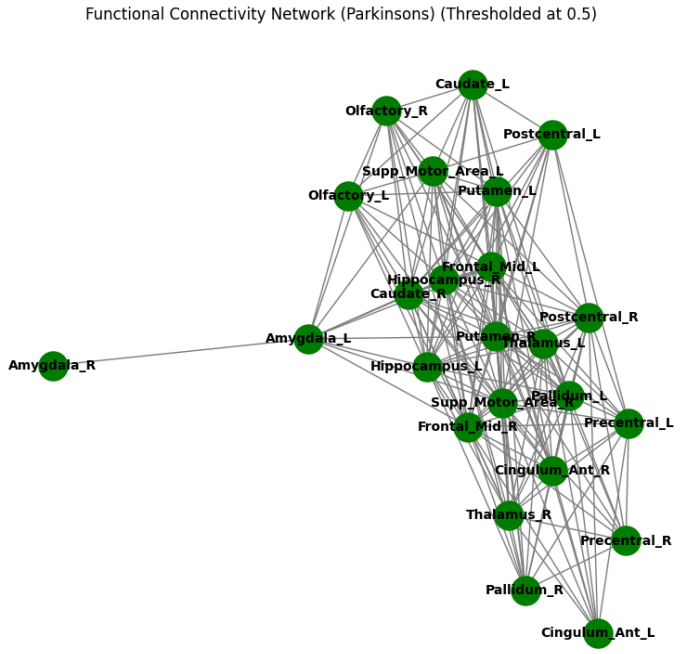



Functional Connectivity (Healthy)
Functional Connectivity (Parkinsons)
Hippocampus_L
Hippocampus_R
Amygdala_L
Amygdala_R
Caudate_L
Caudate_R
Putamen_L
Putamen_R
Thalamus_L
Thalamus_R
Olfactory_L
Olfactory_R
Pallidum_L
Pallidum_R
Precentral_L
Precentral_R
Postcentral_L
Postcentral_R
Cingulum_Ant_L
Cingulum_Ant_R
Frontal_Mid_L
Frontal_Mid_R
Supp_Motor_Area_L
Supp_Motor_Area_R
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8


Mean fMRI Image (Healthy)
L
R
L
R
y=-1
x=0
z=28


Mean fMRI Image (Parkinsons)
L
R
L
R
y=0
x=0
z=22






```
1. Static Analysis
2. Dynamic Analysis
3. Exit
Enter your choice: 2

Found 20 healthy subjects
Found 20 Parkinson's subjects

Average ROI Switches (Healthy)
Hippocampus_L: 104.90
Hippocampus_R: 105.60
Amygdala_L: 104.40
Amygdala_R: 106.40
Caudate_L: 106.75
Caudate_R: 107.00
Putamen_L: 106.60
Putamen_R: 105.35
Thalamus_L: 106.70
Thalamus_R: 104.20
Olfactory_L: 106.30
Olfactory_R: 107.10
Pallidum_L: 105.95
Pallidum_R: 106.45
Precentral_L: 105.20
Precentral_R: 104.00
Postcentral_L: 105.95
Postcentral_R: 103.90
Cingulum_Ant_L: 106.70
Cingulum_Ant_R: 105.30
Frontal_Mid_L: 105.55
Frontal_Mid_R: 104.50
Supp_Motor_Area_L: 105.50
Supp_Motor_Area_R: 105.05
```

```
Average ROI Switches (Parkinsons)
Hippocampus_L: 108.60
Hippocampus_R: 108.25
Amygdala_L: 110.20
Amygdala_R: 108.85
Caudate_L: 109.30
Caudate_R: 109.50
Putamen_L: 108.35
Putamen_R: 109.50
Thalamus_L: 109.30
Thalamus_R: 107.15
Olfactory_L: 109.80
Olfactory_R: 109.55
Pallidum_L: 108.05
Pallidum_R: 110.50
Precentral_L: 109.00
Precentral_R: 108.60
Postcentral_L: 109.05
Postcentral_R: 110.15
Cingulum_Ant_L: 110.55
Cingulum_Ant_R: 108.55
Frontal_Mid_L: 109.05
Frontal_Mid_R: 110.30
Supp_Motor_Area_L: 108.55
Supp_Motor_Area_R: 110.00
```

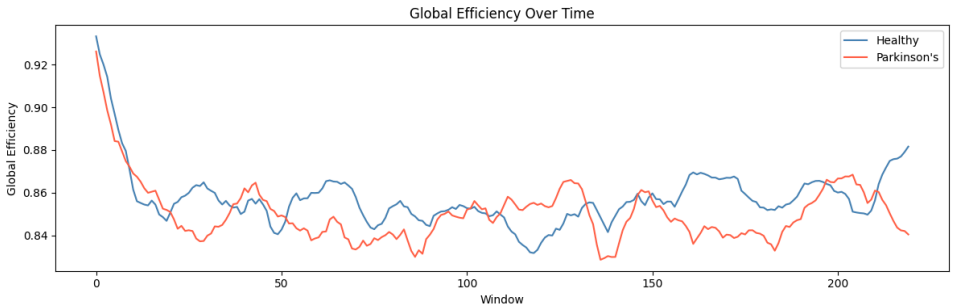
```
Graph Measures Comparison

clustering          Healthy: 0.5072 | Parkinsons: 0.4942
path_length         Healthy: 0.6792 | Parkinsons: 0.6761
global_efficiency   Healthy: 0.8572 | Parkinsons: 0.8505
modularity          Healthy: 0.1215 | Parkinsons: 0.1288
```

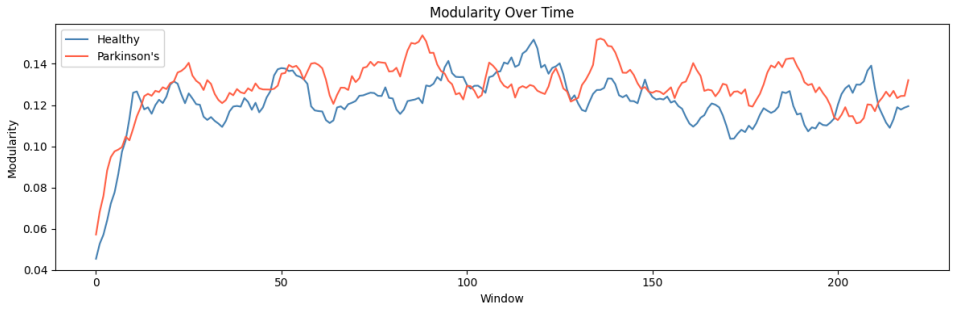
```
Generating animations...
Saved animation to results/animations/healthy_heatmap.gif
Saved individual heatmap frames to results/frames/heatmaps/
Saved animation to results/animations/pd_heatmap.gif
Saved individual heatmap frames to results/frames/heatmaps/
Saved animation to results/animations/healthy_network.gif
Saved individual network frames to results/frames/networks/
Saved animation to results/animations/pd_network.gif
Saved individual network frames to results/frames/networks/
avg_healthy mean: 105.63958333333335
avg_parkinsons mean: 109.19583333333333
ROI switching comparison saved to results/plots/roi_switching_comparison.png
Recurrence plot saved to results/plots/recurrence_plot.png
Report saved to results/report.pdf
```

Node Strength Comparison (average per ROI)

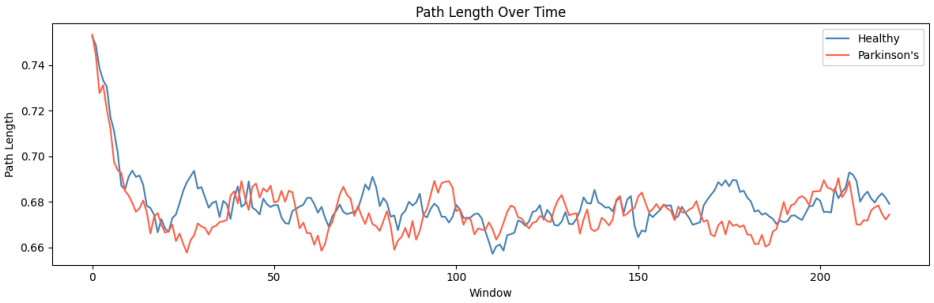
ROI	Healthy	Parkinsons
Hippocampus_L	11.1168	10.8582
Hippocampus_R	10.8579	10.4614
Amygdala_L	9.1056	8.5118
Amygdala_R	8.7999	8.8610
Caudate_L	10.7842	10.7917
Caudate_R	10.9086	10.9458
Putamen_L	11.1393	10.4434
Putamen_R	10.8603	10.8404
Thalamus_L	10.7834	10.0929
Thalamus_R	10.8715	10.2834
Olfactory_L	8.7769	8.8238
Olfactory_R	8.3607	8.3484
Pallidum_L	9.7829	9.5792
Pallidum_R	9.6398	9.6398
Precentral_L	11.5101	10.5582
Precentral_R	10.9856	10.4362
Postcentral_L	10.5852	9.7171
Postcentral_R	10.0466	9.9118
Cingulum_Ant_L	9.3098	9.3684
Cingulum_Ant_R	9.5169	9.7786
Frontal_Mid_L	11.7157	10.8441
Frontal_Mid_R	11.8188	11.2913
Supp_Motor_Area_L	11.4774	10.8398
Supp_Motor_Area_R	11.6344	11.0626

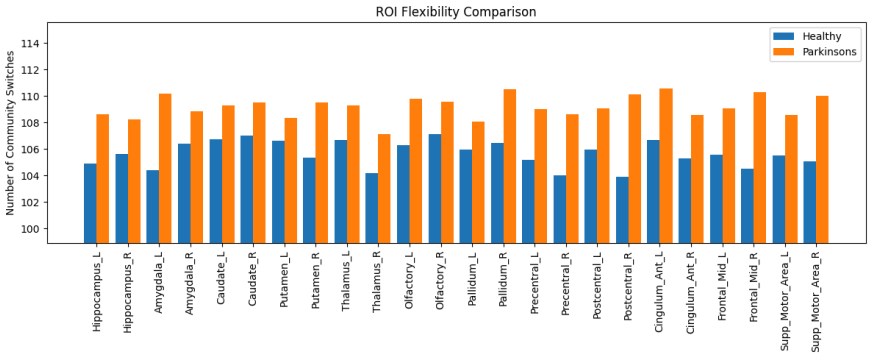
2. Dynamic Analysis Results

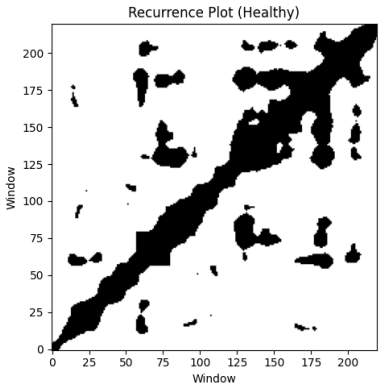


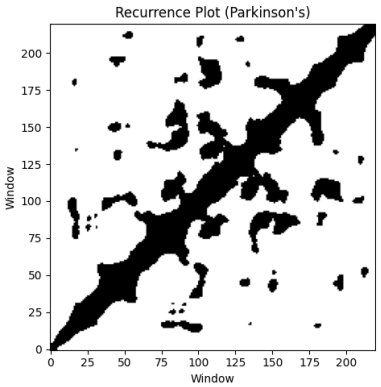


Clustering Over Time
Healthy
Parkinson's
Clustering
0.65
0.60
0.55
0.50
0.45
0
50
100
150
200
Window


Path Length Over Time
Healthy
Parkinson's
Path Length
0.74
0.72
0.70
0.68
0.66
0
50
100
150
200
Window


ROI Flexibility Comparison
Healthy
Parkinsons
Number of Community Switches
114
112
110
108
106
104
102
100
Hippocampus_L
Hippocampus_R
Amygdala_L
Amygdala_R
Caudate_L
Caudate_R
Putamen_L
Putamen_R
Thalamus_L
Thalamus_R
Olfactory_L
Olfactory_R
Pallidum_L
Pallidum_R
Precentral_L
Precentral_R
Postcentral_L
Postcentral_R
Cingulum_Ant_L
Cingulum_Ant_R
Frontal_Mid_L
Frontal_Mid_R
Supp_Motor_Area_L
Supp_Motor_Area_R


Recurrence Plot (Healthy)
Window
200
175
150
125
100
75
50
25
0
0
25
50
75
100
125
150
175
200
Window


Recurrence Plot (Parkinson's)
Window
200
175
150
125
100
75
50
25
0
0
25
50
75
100
125
150
175
200
Window

Conclusion

This project analyzes dynamic functional connectivity in brain networks using resting-state fMRI data.

It applies sliding window analysis and spectral clustering to capture temporal changes in interactions between brain regions.

The approach helps understand alterations in brain network dynamics associated with Parkinson's disease, supporting neuroscience research and brain network analysis.

Future Enhancements

- Extend the system to analyze larger and more diverse fMRI datasets for more reliable insights into brain network dynamics.
- Incorporate advanced machine learning or deep learning models to improve detection of neurological patterns related to Parkinson's disease.
- Develop interactive visualization tools for better interpretation of dynamic brain connectivity patterns.

References

1. Allen, E. A., et al. “Tracking whole-brain connectivity dynamics in the resting state.” Cerebral Cortex, 24(3), 663–676, 2014.
2. von Luxburg, U. “A tutorial on spectral clustering.” Statistics and Computing, 17(4), 395–416, 2007.
3. Bassett, D. S., & Sporns, O. “Network neuroscience.” Nature Neuroscience, 20(3), 353–364, 2017.
4. Rubinov, M., & Sporns, O. “Complex network measures of brain connectivity: Uses and interpretations.” NeuroImage, 52(3), 1059–1069, 2010.
5. OpenNeuro. OpenNeuro Neuroimaging Dataset Repository. Available: https://openneuro.org
6. Python Software Foundation. Python Documentation. Available: https://docs.python.org

Thank You

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
RAJAGIRI SCHOOL OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)
RAJAGIRI VALLEY, KAKKANAD, KOCHI, 682039
(Affiliated to APJ Abdul Kalam Technological University)

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication:Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, lan-

guage, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes

A graduate of the Computer Science and Engineering Program will demonstrate:

CO1

Identify technically and economically feasible problems.

CO2

Identify and survey the relevant literature for getting exposed to related solutions and get

familiarized with software development processes.

CO3

Perform requirement analysis, identify design methodologies and develop adaptable and reusable solutions of minimal complexity by using modern tools and advanced programming techniques.

CO4

Prepare technical report and deliver presentation.

CO5

Apply engineering and management principles to achieve the goal of the project.

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES:

Sl.No	DESCRIPTION	Blooms' Taxonomy Level
CO1	Identify technically and economically feasible problems.	Apply
CO2	Identify and survey the relevant literature for getting exposed to related solutions and get familiarized with software development processes.	Apply
CO3	Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions of minimal complexity by using modern tools & advanced programming techniques.	Apply
CO4	Prepare technical report and deliver presentation.	Apply
CO5	Apply engineering and management principles to achieve the goal of the project.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO 1	PO 2	PO 3	PO 4	PO5	PO6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	2	2	3	2	2	2	3	2	2	2
CO2	3	3	3	3	3	-	3	2	3	2	3	2	2	2
CO3	3	3	3	3	3	2	3	2	2	2	3	-	-	2
CO4	2	3	2	2	2	-	3	3	3	2	3	2	2	2
CO5	3	3	3	2	2	2	3	2	-	2	3	2	2	2

JUSTIFICATIONS FOR CO-PO MAPPING

MAPPING	LOW/MEDIUM /HIGH	JUSTIFICATION
101003/CS622T.CO1-PO1	**HIGH**	Identify technically and economically feasible problems by applying the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
101003/CS622T.CO1-PO2	**HIGH**	Identify technically and economically feasible problems by analysing complex engineering problems reaching substantiated conclusions using first principles of mathematics.
101003/CS622T.CO1-PO3	**HIGH**	Design solutions for complex engineering problems by identifying technically and economically feasible problems
101003/CS622T.CO1-PO4	**HIGH**	Identify technically and economically feasible problems by analysis and interpretation of data
101003/CS622T.CO1-PO5	**MEDIUM**	Responsibilities relevant to the professional engineering practice by identifying the problem.
101003/CS622T.CO1-PO6	**MEDIUM**	Identify technically and economically feasible problems by understanding the impact of the professional engineering solutions
101003/CS622T.CO1-PO7	**HIGH**	Apply ethical principles and commit to professional ethics to identify technically and economically feasible problems
101003/CS622T.CO1-PO8	**MEDIUM**	Identify technically and economically feasible problems by working as a team
101003/CS622T.CO1-PO9	**MEDIUM**	Communicate effectively with the engineering community by identifying technically and economically feasible problems
101003/CS622T.CO1-P010	**MEDIUM**	Demonstrate knowledge and understanding of engineering and management principles by selecting the technically and economically feasible problems
101003/CS622T.CO1-PO11	**HIGH**	Identify technically and economically feasible problems for long term learning.
101003/CS622T.CO1-PSO1	**MEDIUM**	Ability to identify, analyze and design solutions to identify technically and economically feasible problems
101003/CS6	**MEDIUM**	By designing algorithms and applying standard practices in

22T.CO1-PSO2		software project development and identifying technically and economically feasible problems
101003/CS622T.CO1-PSO3	**MEDIUM**	fundamentals of computer science in competitive research can be applied to Identify technically and economically feasible problems
101003/CS622T.CO2-PO1	**HIGH**	Identify and survey the relevant by applying the knowledge of mathematics, science, engineering fundamentals,
101003/CS622T.CO2-PO2	**HIGH**	Identify, formulate, review research literature, and analyze complex engineering problems get familiarized with software development processes
101003/CS622T.CO2-PO3	**HIGH**	Design solutions for complex engineering problems and design based on the relevant literature
101003/CS622T.CO2-PO4	**HIGH**	Use research-based knowledge including design of experiments based on relevant literature
101003/CS622T.CO2-PO5	**HIGH**	Identify and survey the relevant literature for getting exposed to related solutions and get familiarized with software development processes by using modern tools.
101003/CS622T.CO2-PO6	**MEDIUM**	Create, select, and apply appropriate techniques, resources, by identifying and surveying the relevant literature
101003/CS622T.CO2-PO7	**HIGH**	Apply ethical principles and commit to professional ethics based on the relevant literature
101003/CS622T.CO2-PO8	**MEDIUM**	Identify and survey the relevant literature as a team
101003/CS622T.CO2-PO9	**HIGH**	Identify and survey the relevant literature for a good communication to the engineering fraternity.
101003/CS622T.CO2-PO10	**MEDIUM**	Identify and survey the relevant literature to demonstrate knowledge and understanding of engineering and management principles
101003/CS622T.CO2-PO11	**HIGH**	Identify and survey the relevant literature for independent and lifelong learning
101003/CS622T.CO2-PSO1	**MEDIUM**	Design solutions for complex engineering problems by Identifying and survey the relevant literature
101003/CS622T.CO2-	**MEDIUM**	Identify and survey the relevant literature for acquiring programming efficiency by designing algorithms and applying

PSO2		standard practices
101003/CS622T.CO2-PSO3	**MEDIUM**	Identify and survey the relevant literature to apply the fundamentals of computer science in competitive research
101003/CS622T.CO3-PO1	**HIGH**	Perform requirement analysis, identify design methodologies by using modern tools & advanced programming techniques and by applying the knowledge of mathematics, science, engineering fundamentals.
101003/CS622T.CO3-PO2	**HIGH**	Identify, formulate, review research literature for requirement analysis, identify design methodologies and develop adaptable & reusable solutions
101003/CS622T.CO3-PO3	**HIGH**	Design solutions for complex engineering problems and perform requirement analysis, identify design methodologies.
101003/CS622T.CO3-PO4	**HIGH**	Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions
101003/CS622T.CO3-PO5	**HIGH**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools
101003/CS622T.CO3-PO5	**MEDIUM**	Perform requirement analysis, identify design methodologies and assess societal, health, safety, legal, and cultural issues.
101003/CS622T.CO3-PO6	**MEDIUM**	Understand the impact of the professional engineering solutions in societal and environmental contexts and Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions
101003/CS622T.CO3-PO7	**HIGH**	Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions by applying ethical principles and commit to professional ethics
101003/CS622T.CO3-PO8	**MEDIUM**	Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings
101003/CS622T.CO3-PO9	**MEDIUM**	Communicate effectively with the engineering community and with society at large to perform requirement analysis, identify design methodologies
101003/CS622T.CO3-PO10	**MEDIUM**	Demonstrate knowledge and understanding of engineering requirement analysis by identifying design methodologies
101003/CS622T.CO3-PO11	**HIGH**	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change by analysis, identify design

		methodologies and develop adaptable & reusable solutions.
101003/CS622T.CO3-PSO3	**MEDIUM**	The ability to apply the fundamentals of computer science in competitive research and prior to that perform requirement analysis, identify design methodologies
101003/CS622T.CO4-PO1	**MEDIUM**	Prepare technical report and deliver presentation by applying the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
101003/CS622T.CO4-PO2	**HIGH**	Identify, formulate, review research literature, and analyze complex engineering problems by preparing technical report and deliver presentation.
101003/CS622T.CO4-PO3	**MEDIUM**	Prepare Design solutions for complex engineering problems and create technical report and deliver presentation
101003/CS622T.CO4-PO4	**MEDIUM**	Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions and prepare technical report and deliver presentation
101003/CS622T.CO4-PO5	**MEDIUM**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools and prepare technical report and deliver presentation
101003/CS622T.CO4-PO7	**HIGH**	Prepare technical report and deliver presentation by applying ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice
101003/CS622T.CO4-PO8	**HIGH**	Prepare technical report and deliver presentation effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings
101003/CS622T.CO4-PO9	**HIGH**	Communicate effectively with the engineering community and with society at large by prepare technical report and deliver presentation.
101003/CS622T.CO4-PO10	**MEDIUM**	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work by prepare technical report and deliver presentation.
101003/CS622T.CO4-PO11	**HIGH**	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change by prepare technical report and deliver presentation
101003/CS622T.CO4-PSO1	**MEDIUM**	Prepare a technical report and deliver presentation to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas.
101003/CS6	**MEDIUM**	To acquire programming efficiency by designing algorithms

22T.CO4-PSO2		and applying standard practices in software project development and to prepare technical report and deliver presentation
101003/CS622T.CO4-PSO3	**MEDIUM**	To apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs by preparing technical report and deliver presentation
101003/CS622T.CO5-PO1	**HIGH**	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
101003/CS622T.CO5-PO2	**HIGH**	Identify, formulate, review research literature, and analyze complex engineering problems by applying engineering and management principles to achieve the goal of the project.
101003/CS622T.CO5-PO3	**HIGH**	Apply engineering and management principles to achieve the goal of the project and to design solutions for complex engineering problems and design system components or processes that meet the specified needs
101003/CS622T.CO5-PO4	**MEDIUM**	Apply engineering and management principles to achieve the goal of the project and use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions
101003/CS622T.CO5-PO5	**MEDIUM**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools and to apply engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-PO6	**MEDIUM**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities by applying engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-PO7	**MEDIUM**	Understand the impact of the professional engineering solutions in societal and environmental contexts, and apply engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-PO8	**HIGH**	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice and to use the engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-PO9	**MEDIUM**	Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings and to apply engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-	**MEDIUM**	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as

PO11		a member and leader in a team. Manage projects in multidisciplinary environments and to apply engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-PO11	**HIGH**	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change and to apply engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-PSO1	**MEDIUM**	The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas Apply engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-PSO2	**MEDIUM**	The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry and to apply engineering and management principles to achieve the goal of the project
101003/CS622T.CO5-PSO3	**MEDIUM**	The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur and apply engineering and management principles to achieve the goal of the project